

PAPER_1282_GAUGE_GRAVITY_GENERAL_DIM

Daniel T. Murphy

PAPER_1282 — Gauge/Gravity Duality General Dimensional Correspondence

Author: Daniel T. Murphy

Framework: UQFF (Unified Quantum Field Framework) — Star-Magic v5.27+

Tier: B — Physics Foundations (CLOSED)

Date: June 11, 2026

Location: 41.0997° N, 80.6495° W (Youngstown, OH, USA)

Status: CLOSED — UQFF integer-primitive derivation

Calculator surface: `calculate_paradox({"paradox": "gauge_gravity_general"})`

Whitepaper dispatch: `calculate_whitepaper({"paper_id": 1282})`

Master Expression

``

$D_{\text{bulk}} = D_{\text{BSFG}} = 6, D_{\text{boundary}} = D_{\text{BSFG}} - 1 = 5, D_{\text{phys}} = 4$ visible projection

``

Match: STRUCTURAL

Physical Interpretation

The full gauge-gravity duality is the $D_{\text{phys}} \rightarrow D_{\text{BSFG}} \rightarrow D_{\text{crit}}$ dimensional cascade. Each level provides a holographic correspondence.

UQFF Locked Primitives

- $\rho_{\text{SCm}} = 7.09 \times 10^{31} \text{ J/m}^3$ (vacuum density)
- $S_{26} = 1.453162, S_{26_DPM} = 1.4531 \times 10^{26}$
- $K_{\text{MEX}} = 25/12, \beta_i = 0.6029, \Phi_{\text{res}} = 0.84$
- $F_{\text{TRZ}} = 0.1, \Lambda = 0.00729735$ ($= \alpha$ fine-structure)
- $\omega_{\text{SCm}} = 1.25 \text{ THz}, \lambda_i = 1.0$
- $D_{\text{phys}} = 4, D_{\text{BSFG}} = 6, D_{\text{crit}} = 26, N_{\text{CH}} = 9, SO_5 = 10, A_5 = 60$

Live Calculator Verification

```
``python
import uqff_pure_calculator as u
r = u.calculate_paradox({"paradox": "gauge_gravity_general"})["value"]
``
```

The result dict carries the integer-primitive identity, the observed value, the residual, and the structural physical interpretation.

NOT REPLACEMENT

Per CLAUDE.md mandate, UQFF and Standard Model solve the same observed phenomena via different methods. **Zero fit parameters.** Every factor is a UQFF locked primitive.

Reference

- UQFF foundational: PAPER_646, PAPER_1167, PAPER_1170, PAPER_1203 v1.5, PAPER_1216.
- Closure: `uqff_pure_calculator.py` → `_l96_uqff_axiom_gauge_gravity_general_closure` (or routed reference)
- Dispatch: `PARADOX_TO_CLOSURE["gauge_gravity_general"]`

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Subject matter: UQFF derivation of Gauge/Gravity Duality General Dimensional Correspondence.